

Enumerations in Java

* Understanding Enum

```
enum Direction {  
    NORTH,  
    SOUTH,  
    EAST,  
    WEST  
}
```

compile →

```
class Direction extends Enum<Direction> {  
}
```

- java.lang.Enum overrides ✓
- each constants are static & final
- & are objects of class Direction

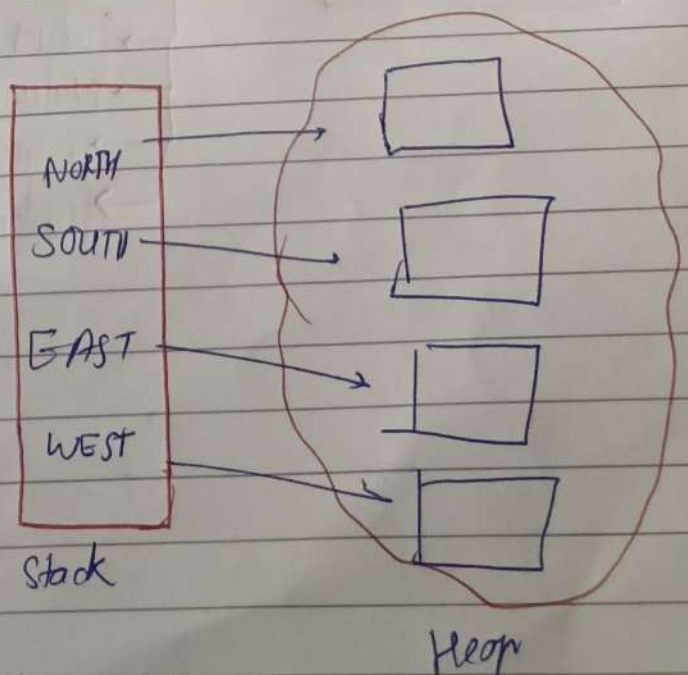
```
public final class Direction extends Enum<Direction> {  
    public static final Direction NORTH = new Direction();  
    public static final Direction SOUTH = new Direction();  
    public static final Direction EAST = new Direction();  
    public static final Direction WEST = new Direction();  
}
```

```
private Direction() {}
```

```
main() {  
    ✗ Direction d = new Direction();  
    not possible  
    as constructor is  
    private
```

```
✓ Direction d = Direction.NORTH;
```

```
}
```

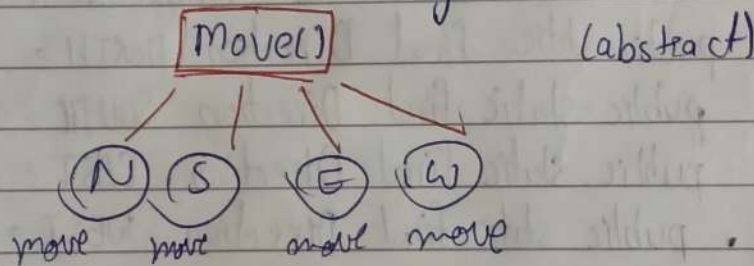
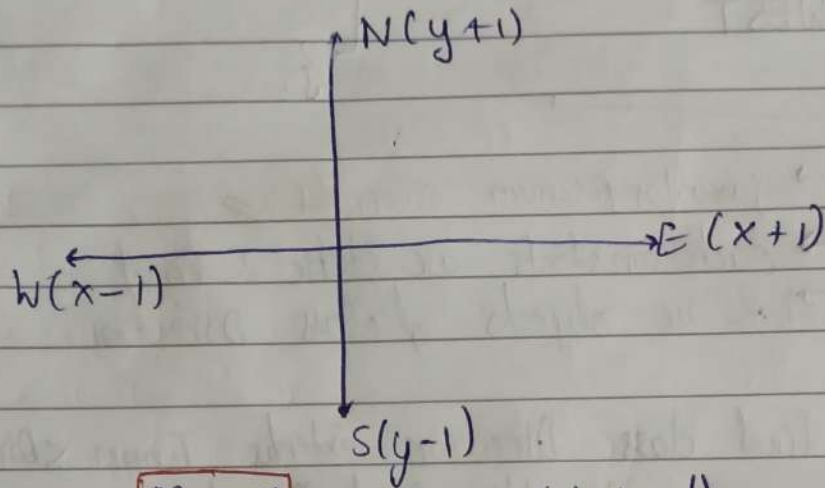
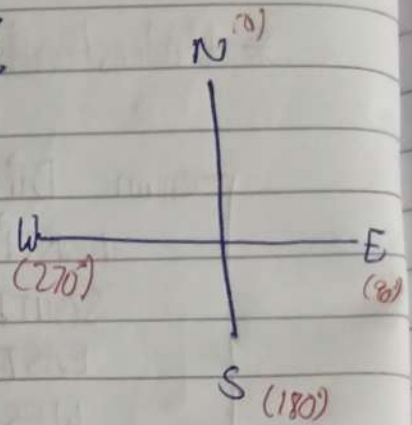


* Enum $\rightarrow$ Class

$\rightarrow$ States
 $\rightarrow$ Behaviour

enum Direction {

NORTH,
SOUTH,
EAST,
WEST;



enum Direction

Enum (java.lang)

values()
valueOf(String)
name()
ordinal()

$\rightarrow$ compiler generated

Enum class

More in depth

```
enum Direction {  
    NORTH,  
    SOUTH,  
    EAST,  
    WEST  
}
```

final class Direction extends Enum<Direction> {


```
    private static final Direction[] $VALUES = {  
        NORTH,  
        SOUTH,  
        EAST,  
        WEST;  
    }
```

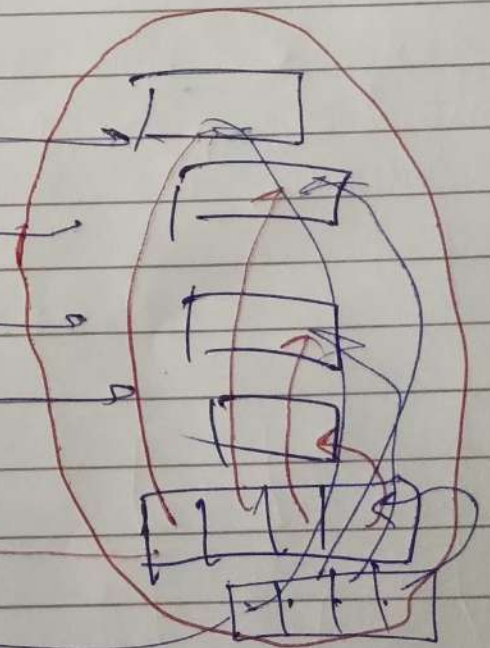
Compiler
generated
methods

```
    public static Direction[] values() {  
        return $VALUES.clone()  
    }
```

}

NORTH
SOUTH
EAST
WEST
\$VALUES
[] of

Stack



Heap

*valueOf()
 → compile generated
 → but also in Enum class
 → but we use mostly compile generated method

public static Direction valueOf(String s)
[3 super.valueOf (Direction.class(), s);

* Enums

- └ Day of the week
- └ Payment status
- └ Order status
- └ logs (INFO, ERROR, DEBUG)